UNIVERSITY OF ABOMEY CALAVI (UAC)

DOCTORAL SCHOOL OF AGRONOMIC SCIENCES AND WATER

Specialty: Master in Biostatistics

Type of exam: Artificial Intelligence

Duration: 1h30

Exercise 1: Multiple-choice questions. (15pts)

Policy: For each question/sentence below, select all letters with correct phrases/words (several possible choices). Don't write the sentence, you just need to specify the letters. For example, for question 8, an answer could be "8. a, c, d". Put "None" (and explain) if you think no letter corresponds to something right. Justification is not necessary but it is recommended to abstain if your choices do not convince you. [1 pt for good answer, -1pt for bad answer, 0pt for no answer.]

1. **What is the difference between supervised and unsupervised learning?**

a) Supervised learning uses unlabeled data.
b) Unsupervised learning does not use training data.
c) Supervised learning uses tagged data to train the model.

2. **What would you use as a classification instead of a regression?**

   a) Predict the price of a house.
   b) Categorize emails as spam or non-spam.
   c) Identify the main themes in a set of documents.

3. **What is overfitting in the context of machine learning?**

a) The model is too simple for training data.
b) The model adapts too much to training data and does not generalize well.
c) The model fails to adjust to the training data.

4. **What is a confusion matrix in the context of classification?**

a) A matrix that indicates whether the model is confused or not.
b) A matrix that summarizes the model's performance by comparing predictions to true labels.
c) A matrix that shows the correlation between the different characteristics.

5. **What is the difference between the validation set and the test set?**

a) The validation assembly is used during training, the test assembly is used after training.
b) The test set is used during training, and the validation set is used after training.
c) Both sets are used during training

6. What is the main disadvantage of the K-means algorithm?

a) Sensitive to cluster center initialization
b) Only works for linearly separable data
c) Cannot be used for large data
(d) Cannot be used for multi-class classification

7. How does the Apriori algorithm work to generate association rules?

a) Identifying frequent sets of items from a transactional database
(b) Using regression techniques to predict associations between variables
c) Applying clustering algorithms to the data
(d) None of the above

8. What is the impact of choosing a higher confidence threshold in the Apriori algorithm?

a) Generates fewer but more reliable rules
b) Generates more rules but is less reliable
(c) No impact on the number or reliability of rules generated
(d) None of the above

9. How to interpret trust in an association rule?

(a) The likelihood that the antecedent will result in the
(b) The likelihood that the consequence will result in the antecedent
c) The number of times the rule occurs in the database
(d) The frequency of the history in the database

10. What is a hidden layer in a neural network?

a) A layer that contains the network inputs
b) A layer that contains the network outputs
(c) A layer that does not communicate directly with the input or output
(d) None of the above

11. What is the activation function in a neural network?

a) A function that standardizes network inputs
b) A function that transforms the weighted sum of inputs into a non-linear output
(c) A function that reduces the dimensions of the entrance

12. What is the difference between a global constraint and a local constraint in constraint programming?

a) A global constraint specifies a relationship between all variables, while a local constraint specifies a relationship between only two variables

(b) A global constraint may or may not be satisfied, while a local constraint must always be satisfied

(c) A global constraint can be expressed mathematically, while a local constraint cannot

(d) None of the above

13. How does the minimax algorithm work in adversarial search?

a) It explores the game tree by alternating between the max and min player levels, always choosing the optimal movement for each player

b) Explores the game tree by randomly selecting the movements at each level

c) It seeks to minimize the total number of nodes explored in the game tree

(d) None of the above

14. How can the measure of equity be quantified in AI models?

a) Using clustering techniques to group data based on sensitive characteristics.

(b) Calculating differences in model performance between different demographic or category groups.

(c) Evaluating the overall accuracy of the model over all data.

(d) None of the above.

15. How can representation bias be detected in datasets?

(a) Examining the proportion of each demographic or category in the data set.

(b) Evaluating the overall accuracy of the model over all data.

(c) Comparing model performance across different demographic or category groups.

(d) None of the above.

Exercise 2 (5pts)

Data on minimum temperature, maximum temperature, insolation, and rainfall have been collected over the last 26 years to predict soybean yield in the municipality of Bohicon. 75% of the raw data is used to train the models, and 25% to test the different models. Three regression models are used for this prediction. These are polynomial multiple regression, simple linear regression, and artificial neural networks. The Root Mean Square Error (RMSE) is the metric used to compare the predicted values with the actual values of the different models. Thus, the values obtained are 5.36 e-16 for the polynomial regression, 0.01 for the linear regression, and 0.002 for the artificial neural networks.

1. What step did you miss before partitioning the data?
2. What type of machine learning is it?
3. What is the ideal value of the RMSE for the model to be accurate?
4. Provide a good interpretation of each model.
5. What can you conclude after a comparison of the different models?